Day 6 — Assignment

More Numpy & Pandas Practice — from scratch

Like the notebooks today, there's no starter code here — write every cell yourself. Target time: 30–45 minutes. Section 1 uses plain arrays; Section 2 uses winequality-red.csv, winequality-white.csv, and bike_share_data.csv.

Section 1 — Numpy From Scratch

Task 1: Create these four arrays: [2, 5, 8, 11, 14] from a list; the integers 2 through 20 using arange(); 0 to 5 in steps of 0.4; and a 4x2 array of random floats.

Task 2: Create an array from the tuple (4, '6', 8, '11', 2, '9'), casting it to integers. Check its dtype and shape.

Task 3: Update the 3rd value (index 2) of that array to 20.

Task 4: Reshape the array into 2 rows and 3 columns.

Task 5: Add 5 to every value in row 1, and add 10 to every value in row 2.

Task 6: Multiply column 1 by 2, column 2 by 3, and column 3 by 4 (all in one broadcasting operation).

Task 7: Find the sum of each row, and the maximum of each column.

Task 8: Find the index position of the value 42 in the array (it should appear exactly once).

Task 9: More challenges, each independent: (a) build an array with at least two NaN values mixed in, and test which elements are NaN; (b) build one array containing 8 twos, 8 nines, and 8 ones, all concatenated together; (c) build a 3x3 array and iterate over every element with np.nditer(); (d) build a 2x2x2 array of random values and confirm its shape; (e) add the vector [1, 2, 3] to every row of a 2x3 matrix of your choice using broadcasting.

Section 2 — Combining, Binning, and Pivoting

Task 10: Read both wine datasets. Find the mean sulphates for red and white separately. For each type, build a flag column marking which wines are ABOVE their own type's mean sulphates, attach it using both .join() and pd.concat(), then return just the rows that are AT OR BELOW the mean.

Task 11: Bin red wine's pH column into 4 equal-width bins. Attach the bin labels to the DataFrame, then return every row that is NOT in the highest pH bin.

Task 12: Using only the wines at or below their type's mean sulphates (from Task 10), build a pivot table showing average sulphates by quality, separately for red and white.

Task 13: Read bike_share_data.csv and clean its column names (lowercase, underscores). Create a duration_group column using pd.cut() with these bin edges: 0, 300, 600, 1200, 3600, 300000 seconds, labeled '0-5min', '5-10min', '10-20min', '20-60min', '60min+'. Build a pivot table with average duration, using BOTH subscription_type and duration_group together as the index.